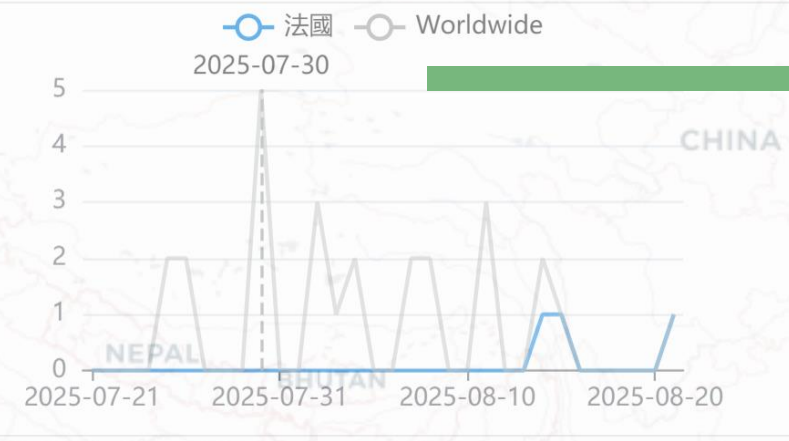




Taiwan CDC's approach to international outbreak monitoring

Epidemic Intelligence Center, Taiwan CDC

10 Sep 2025

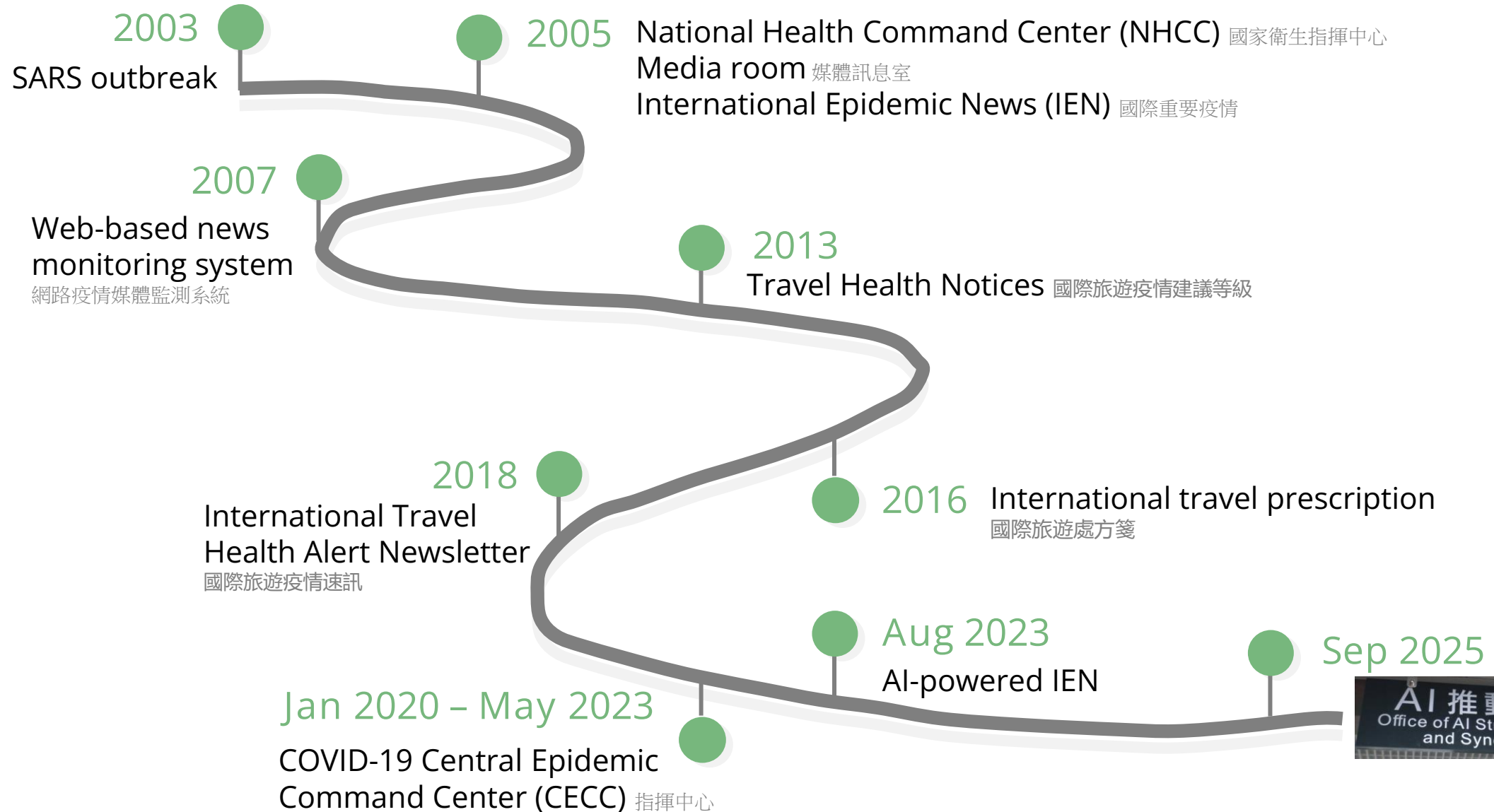


Outline

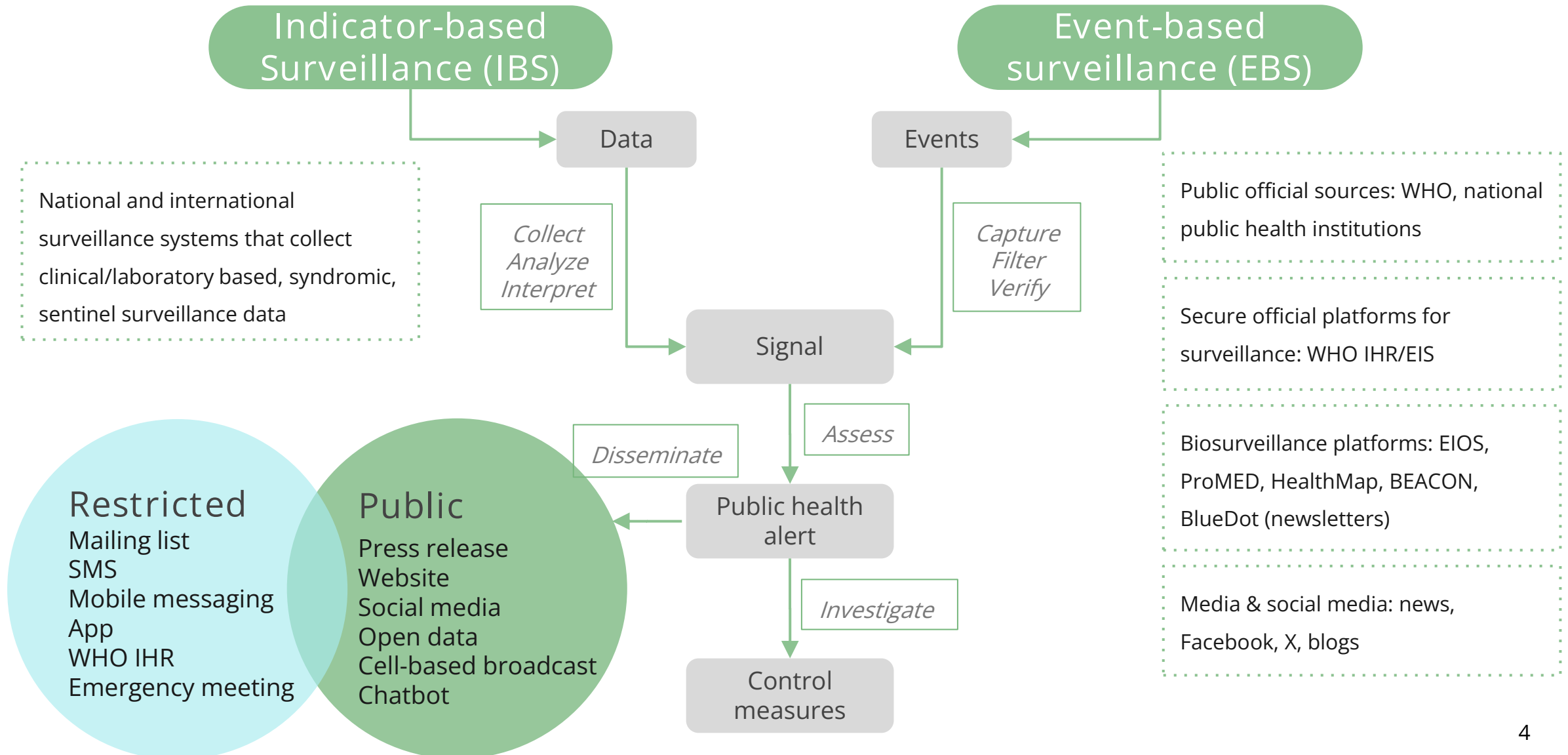
1. How we work
2. IEN in numbers
3. Travel health notices
4. Risk assessment



Milestones



The process of international epidemic intelligence



Workflow



Sources

IBS, EBS, WHO EIS, IHR NFP, OpenAI GPT summary, Bluedot, media monitoring, bilateral information exchange



Data processing

manual data scrapping, Python crawler & ETL, database management, analytics, ChatGPT for writing IEN



Disseminations

email, slides, reports, website updates, press release, datasets, dashboard updates, IHR NFP coordination

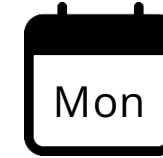
International Epidemic News (IEN)

新加坡/韓國-茲卡病毒感染症

資料來源：outbreaknewstoday、CDA、韓國CDC，2025/6/24

- 新加坡6/18報告2例本土病例，今年迄今累計7例病例，今年病例數與去年同期相當，當局已加強相關監測措施。
- 韓國今年迄今累計2例病例，均為境外移入，該國自2016年以來累計報告41例病例，其中40例為境外移入。

IEN in morning meetings



Mon

Disease in focus



Tue

Vector-borne



Wed

Contact transmitted



Thu

Airborne & aerosol



Fri

Food & water-borne

IEN in numbers

2009-2025 in average

108

kinds of diseases
tracked annually

159

countries/territories
covered annually

84

country-disease reported
every week

16%

IENs highlighted public
health interventions

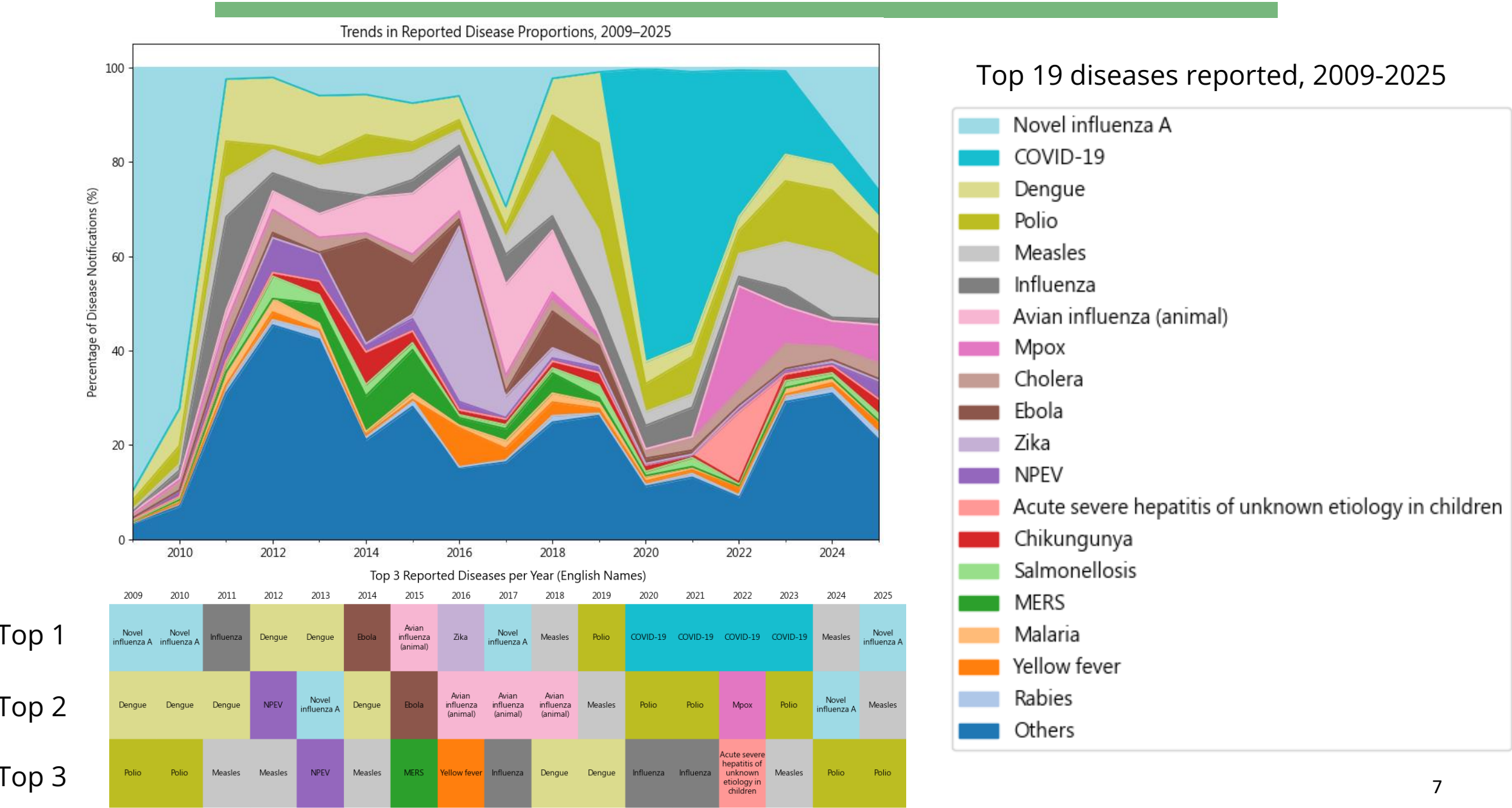
3 days

source-to-publication
interval

481

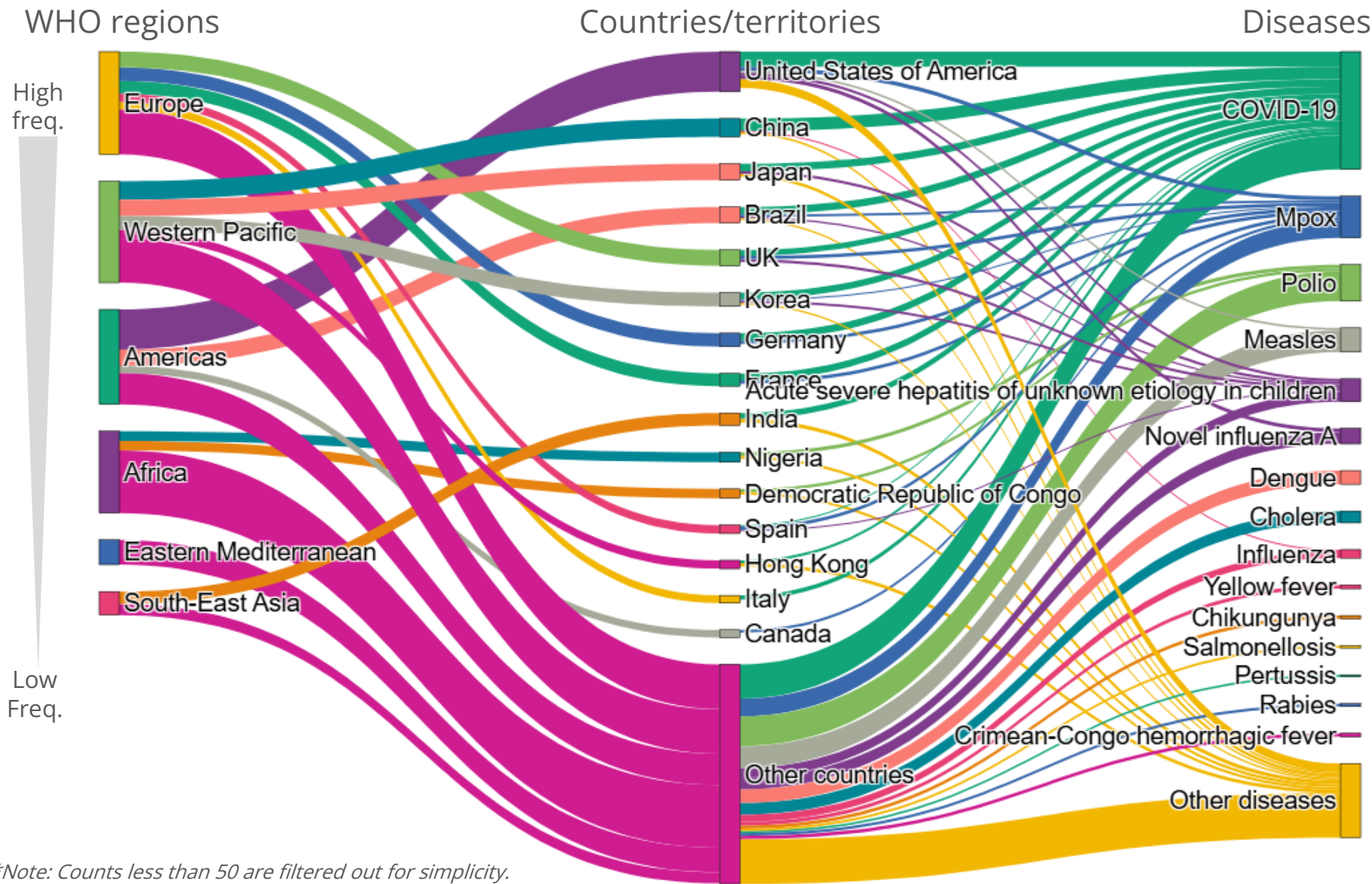
unique sources used
annually *

IEN’s focus has been closely linked to domestic & global outbreaks



IEN's geographical & disease news distributions

International Epidemic News by region, country, and disease, 2021–2025



- Top 5 countries covered in IEN are the US, China, Japan, Brazil, UK
- Top 5 diseases reported for IEN are COVID-19, Mpox, Polio, Measles, Acute severe hepatitis of unknown etiology in children

**Note: Counts less than 50 are filtered out for simplicity.*

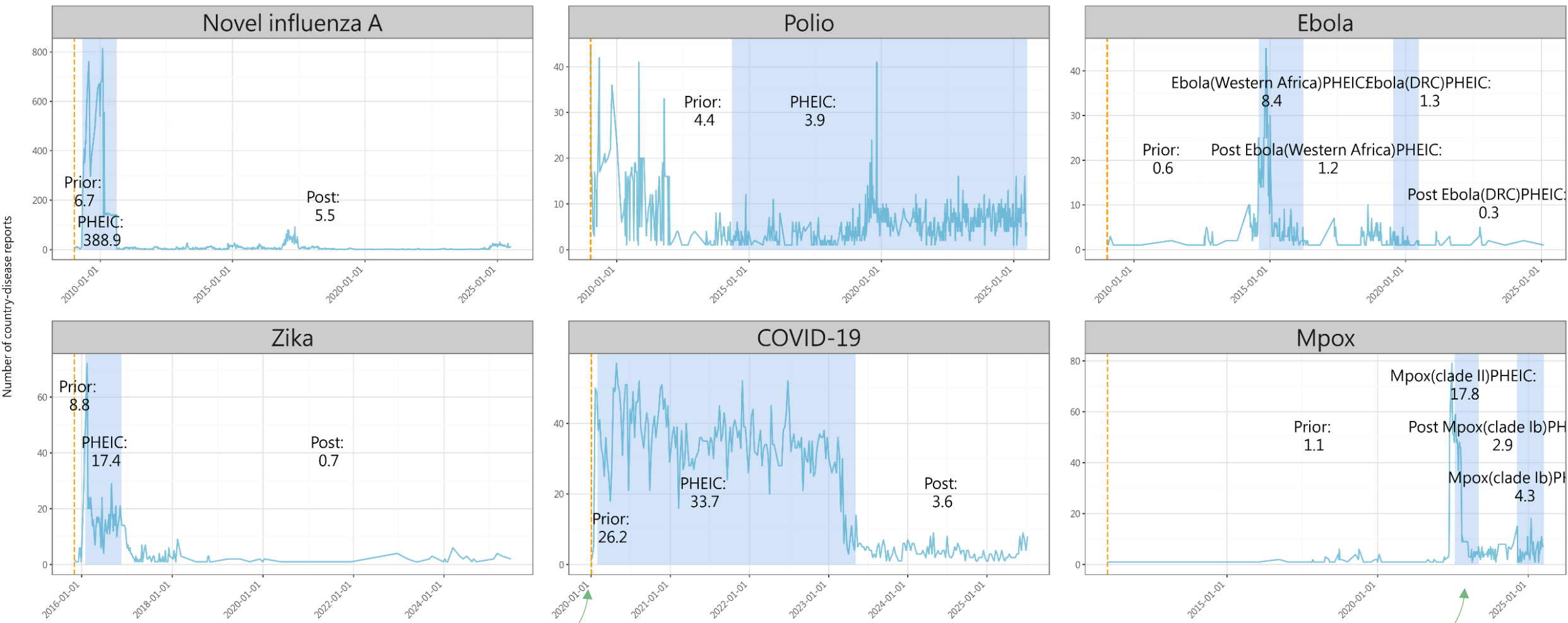
Early alert & enhanced monitoring during WHO PHEICs

Weekly PHEIC Global Epidemic News

Annotated with average weekly counts for the pre-PHEIC, PHEIC, and post-PHEIC phases

First IEN reporting

WHO PHEICs periods



Taiwan CDC first alerted WHO to the Wuhan outbreak on Dec 31, 2019

Except for polio, PHEIC periods saw the highest weekly IEN reports

Travel health notices

19 diseases in focus

- Chikungunya
- Cholera
- COVID-19
- Dengue
- Ebola
- Influenza
- Japanese Encephalitis
- Lassa Fever
- Malaria
- Marburg Virus
- Measles — example →
- MERS-CoV
- Mpox
- Novel Influenza A
- NPEV
- Polio
- SFTS
- Yellow Fever
- Zika

Criteria for level 1, Measles

- Countries reporting more than 1,000 cases in a year, more than 500 cases in 6 months, or more than 250 cases in 3 months
- Countries from which Taiwan has imported cases in 3 months

Criteria for level 2, Measles

- Countries reporting more than 2,500 cases in 3 months
- Countries facing large-scale outbreaks

Travel Health Notices

Types of Notices	Travel Advice
Level 1: Watch	Practice Usual Precautions
Level 2: Alert	Practice Enhanced Precautions
Level 3: Warning	Avoid Nonessential Travel



Quarterly reexamination
of countries' notice levels

AI-driven intelligence pipeline

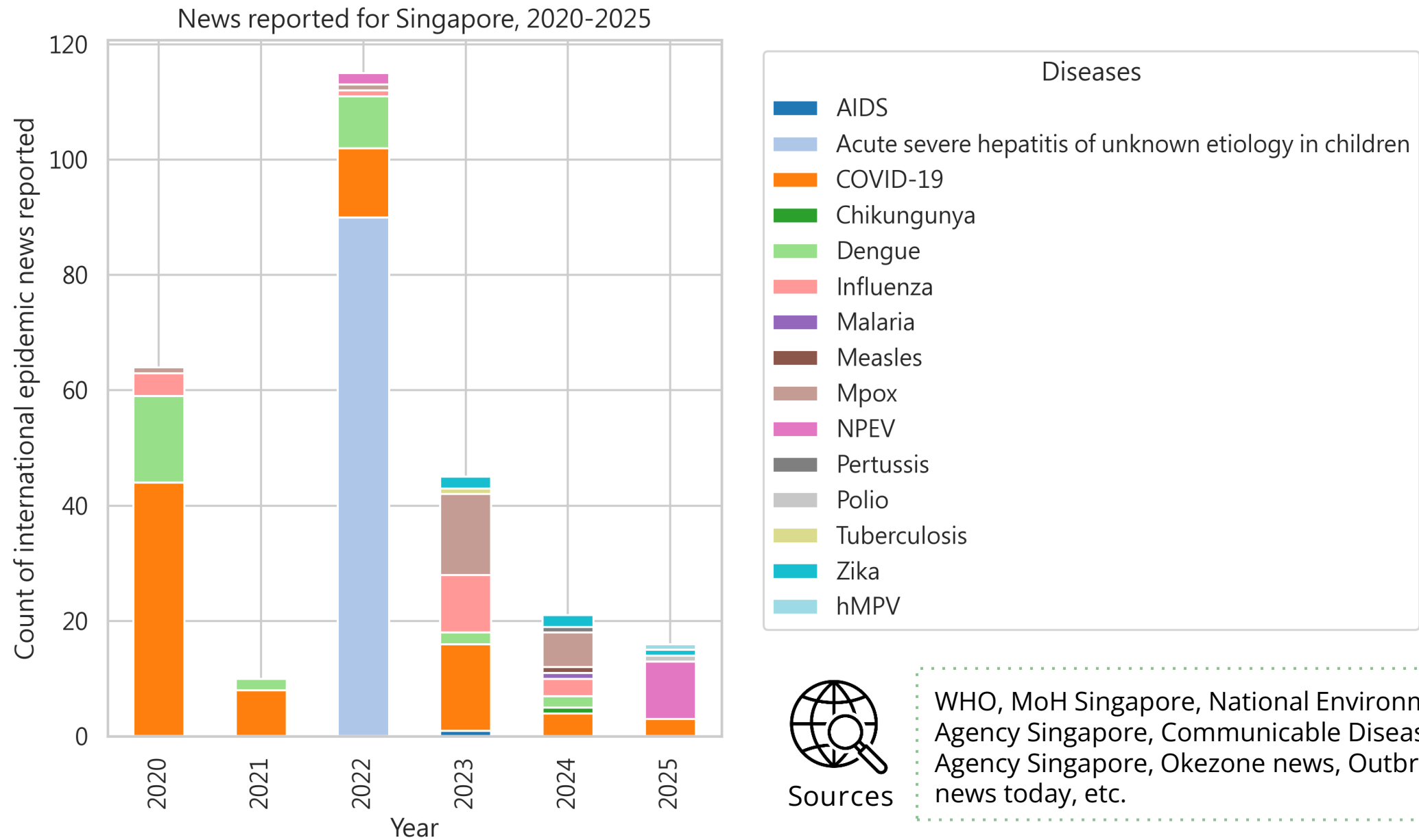
2025-09_news_analysis_gpt_report						
檔案 編輯 查看 插入 格式 資料 工具 擴充功能 說明						
100% 123 預設 (...)						
I11						
	B	C	D	E	F	G
1	process time	publish time	source	disease	country	summary
10	2025-09-03 9:08:3	2025-09-02 8:00:0	Beacon	Highly Pathogeni	Bolivia	傳染病名稱：高病原性禽流感（H5亞型） 國家：玻利維亞 摘要：玻利維亞在聖克魯斯省庫沃市的家禽中確認HPAI H5疫情，已宣布衛生緊急，實施10公里範圍內的控制、撲殺和監測措施，以防止疫情擴散並保障公共健康。
11	2025-09-03 3:34:5	2025-09-02 21:26:1	RelifWeb	Cholera	Democratic Repu	傳染病名稱：霍亂 國家：剛果民主共和國 摘要：剛果民主共和國2025年1-7月報告超過44,800例霍亂病例，死亡率2.8%，超過WHO標準。疫情由於衝突、流離失所、洪水及多重疫情影響，已擴散至18個省份，主要因安全水、衛生設施不足所致。
12	2025-09-03 3:34:5	2025-09-02 21:26:0	RelifWeb	Cholera	Africa	霍亂（霍亂）；非洲；2025年上半年非洲多國爆發霍亂，病例數由2014年的105,287增至2024年的254,075，死亡人數由1,882增至4,725，顯示疫情嚴重攀升。
13	2025-09-03 3:34:5	2025-09-02 20:55:0	CIDRAP	polio	Papua New Guine	傳染病名稱：脊髓灰質炎（小兒麻痺症） 國家：巴布亞新幾內亞 摘要：巴布亞新幾內亞確認首例人類脊髓灰質炎病例，4歲男童因接種疫苗不足感染疫苗衍生病毒，WHO正推行免疫接種以防止疫情擴散。
14	2025-09-02 21:34	2025-09-02 19:34:0	Avian Flu Diary	Influenza A (H1N	Denmark UK	流感病毒（流行性感感冒病毒），英國 摘要：英國因應嚴重流感季節，擴大疫苗資格，放寬抗病毒藥物使用限制，以減輕醫療系統壓力。近期澳洲、丹麥亦出現嚴重流感疫情，且禽流感增加人類感染風險。

- From 10 to 20 news automatically processed on a daily basis
- Powered by OpenAI GPT-4.1-nano model
- Collaborated with Acer Value Lab

Risk assessment

Items	Assessment	Result
Risk of Imported Cases	• Geographical distributions • # of travelers, migrants, trade links • Incubation & latent periods • Screening & border measures • Imported cases historically	Low / medium / high
Risk of Community Transmission	• Transmission route • Local ecology • R0 • Diagnostics & surveillance sensitivity • Community immunity • Interfaces & conditions	Low / medium / high
Impact	• Affected papulation • Vaccine or treatment available • Clinical severity spectrum • Burden on health systems • Economic & social costs • Interaction with other health priorities	Low / medium / high

Singapore in spotlight



Acknowledgement: Data processing & global team, EIC



Thank you

